

Question B — The Bias of the Independence Assumption Between Trees

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The M trees of the forest share the same data stream — in which direction does this bias the article’s formula, which assumes the τ_i independent to compute $\tau_{\text{ARF}} = \min_i \tau_i$? New objects relative to Question A: S is the stream trajectory (x_t, y_t) alone, without the trees’ internal draws (in the campaign a single seed drives both, so G_S cannot be estimated across seeds); U_i the randomness specific to tree i , identified in Section 1; $F(s)$ the marginal, unconditional law of a single tree; $G_S(s) := P(\tau_1 > s \mid S)$ its survival function *given* the stream; $M = 10$ the number of trees in the R2 campaign; β a reliability target, the share of drifts one wants the external monitor to catch ($\beta = 0.95$ in the manuscript).

1 Conditional Factorization

Once S is fixed (the entire trajectory (x_t, y_t) known), what still distinguishes each tree is U_i : the Poisson weights drawn online for bagging (Oza & Russell, 2001 — this is not classical bagging by resampling a fixed dataset, which is impossible in streaming, but a random weight assigned to each new observation), and the random feature subspaces tested at each node.

The first mechanism is the one the manuscript itself invokes: “the M trees in the ARF are trained on the *same* data stream via Poisson-weighted bootstrap (Oza bagging with $\lambda = 6$)” (manuscript, Section III-G, citing Gomes et al. 2017). The second is a defining property of the Adaptive Random Forest in the same reference¹. These draws are, by construction, independent across trees: tree i never sees tree j ’s weights, and a tree newly created at replacement inherits no information (neither weights nor structure) from the tree it replaces — the only link between the two is temporal, the new one having started learning more recently.

The adaptation time of tree i is a deterministic function of the pair (S, U_i) . Once S is fixed, only the variation in U_i remains, and the U_i are i.i.d. across trees. Hence, conditionally on S , the τ_i are indeed independent (and identically distributed, since all trees share the same procedure).

Result. Defining the conditional survival function $G_S(s) := P(\tau_1 > s \mid S)$, conditional independence gives

$$P(\tau_{\text{ARF}} > s \mid S) = P(\forall i, \tau_i > s \mid S) = \prod_{i=1}^M P(\tau_i > s \mid S) = G_S(s)^M.$$

2 The Direction of the Bias, via Convexity

We uncondition using the law of total probability:

$$P(\tau_{\text{ARF}} > s) = \mathbb{E}_S[G_S(s)^M].$$

The function $x \mapsto x^M$ is convex on $[0, 1]$ for $M \geq 1$. Jensen’s inequality gives directly

$$\mathbb{E}_S[G_S(s)^M] \geq (\mathbb{E}_S[G_S(s)])^M = (1 - F(s))^M,$$

since $\mathbb{E}_S[G_S(s)] = P(\tau_1 > s) = 1 - F(s)$ (the marginal law, unconditioned).

¹On the R2 stream it contributes very little, though: the stream carries two features only (x_0, x_1) , so a random subspace selects roughly one of two at each node. Inter-tree diversity there rests almost entirely on the Poisson weights, which suffices for what follows — only the i.i.d. character of the U_i is needed. The campaign leaves `max_features` at `river`’s default, which should be read off the environment that produced the runs rather than assumed.

Result.

$$P(\tau_{\text{ARF}} > s) \geq (1 - F(s))^M \iff P(\tau_{\text{ARF}} \leq s) \leq 1 - (1 - F(s))^M.$$

The true probability of having already adapted by time s is always less than or equal to what the independence formula predicts: the manuscript **overestimates the repair speed** — τ_{ARF} is stochastically larger (slower) than the article claims.

Equality condition. Jensen’s inequality is an equality if and only if $G_S(s)$ is almost surely constant — that is, if the difficulty of adaptation does not actually depend on the specific stream realization. Physically, this case essentially never occurs here: every seed produces a different noise trajectory after the drift, so $G_S(s)$ genuinely varies across realizations. Independence is therefore never exact — only an approximation, systematically optimistic.

3 Operational Consequence

Section 2 fixed the *sign* of the error, and two quantities are affected in opposite directions. The words “optimistic” and “pessimistic” below refer to different subjects, which is what makes the pair look contradictory at first sight.

Quantity	What the independence formula gets wrong
τ_{ARF} , the repair speed	it predicts a <i>faster</i> repair than the real one
P_{miss} , the blind-spot risk	it predicts a <i>larger</i> risk than the real one

The two are consistent, and one implies the other: a repair that is in truth slower leaves the external detector more time to accumulate, so the forest wins the race less often than the formula claims. Overstating the repair speed therefore over-states the blind-spot risk. The manuscript is thus too flattering about the classifier and too alarming about the architecture — and our correction, read alone, makes the blind spot *less* severe than the article reports.

Consequence for M_{crit} . The manuscript’s Corollary 10 turns Proposition 9 into a prescription on ensemble size, and answers this question: *how many trees may the forest carry before the external monitor stops being reliable?* It is evaluated at $\tau_{\text{det}}^* = \lambda/(\Delta e - \delta_P)$, the nominal accumulation time that Proposition 9 substitutes for the random τ_{det} — a substitution Question A examines separately, and which we take as given here.

Requiring $P_{\text{miss}} = 1 - [1 - F(\tau_{\text{det}}^*)]^M \leq 1 - \beta$ gives $[1 - F]^M \geq \beta$, hence $M \ln(1 - F) \geq \ln \beta$, and since $\ln(1 - F) < 0$ the division flips the inequality:

$$M \leq M_{\text{crit}} := \left\lceil \frac{\ln \beta}{\ln[1 - F(\tau_{\text{det}}^*)]} \right\rceil.$$

M_{crit} is therefore a *ceiling* on the number of trees, not a floor — which is the physically sensible reading, since more trees mean a faster repair and hence a worse position for the detector. Adding trees can never help it.

Section 2 established that the true CDF of τ_{ARF} lies everywhere below the independence CDF, so the true P_{miss} is *smaller* than the formula’s at every M . Whatever M the independence certificate admits, the real forest satisfies the guarantee too.

Result. The corollary is not invalidated but **reclassified as a conservative ceiling**: “at most M_{crit} trees” remains a safe prescription, and the true admissible ensemble size is $\geq M_{\text{crit}}$. It is a valid upper limit, no longer a tight one.

One assumption left tacit. The comparison above holds at a fixed τ_{det}^* : it opposes two laws for τ_{ARF} against the same detector behaviour, which is legitimate because Proposition 9 makes $\tau_{\text{det}}^* = \lambda/(\Delta e - \delta_P)$ a constant. Read as a claim about the *real* blind-spot risk, the table above assumes more, since the real τ_{det} is computed on the *ensemble’s* error stream and therefore depends on how correlated the trees are. We do not establish that its law is insensitive to that dependence.

Both effects in fact lower P_{miss} , so the direction of Section 2 is reinforced rather than threatened. A slower repair leaves the excess error in place longer, so the monitor accumulates more and alarms earlier; and correlation weakens bagging’s variance reduction, leaving a burstier error stream that breaches λ on noise alone — the *stochastic resonance* the manuscript reports for a single tree, which the forest normally suppresses. What weakens is not the sign but the *meaning* of P_{miss} : part of the decrease is bought with alarms that win the race without detecting anything. P_{miss} remains a well-defined probability, and a poorer proxy for monitoring quality than the formula suggests.

And the corollary is not rescued either. On the operative grid the ceiling is already empty, which our loosening cannot fill. At $\Delta e = 0.33$ the manuscript measures $\widehat{F}(\tau_{\text{det}}^*) = 0.49$: a *single* tree has a 49% chance of swapping before the monitor’s nominal accumulation time. One tree alone therefore gives $P_{\text{miss}} = 1 - (1 - 0.49)^1 = 0.49$, ten times the 5% the target $\beta = 0.95$ allows, and further trees can only make it worse. No ensemble size meets the guarantee — whence $M_{\text{crit}} = 0$, a ceiling of zero trees.

Our correction states that the true ceiling is $\geq M_{\text{crit}}$, which here reads “ ≥ 0 ” and says nothing: rescuing the certificate would require showing the true ceiling to be ≥ 1 , and a one-sided inequality does not deliver that. The correction therefore changes the corollary’s *status* — an exact figure becomes a conservative limit — without changing its verdict where the manuscript applies it.

4 The Correlation Trap

Toy case $M = 2$ on $\{1, 2\}$: is an anti-monotone coupling compatible with the real ARF?

The counterexample. Support $\{1, 2\}$, marginals $P(\tau_i = 1) = \frac{1}{2}$. Anti-monotone coupling: $\tau_1 = 1 \iff \tau_2 = 2$ (perfectly opposite). Under this coupling, $\{\tau_1 = 2\}$ implies $\{\tau_2 = 1\}$, so $P(\tau_1 > 1, \tau_2 > 1) = P(\tau_1 = 2, \tau_2 = 2) = 0$, hence

$$\text{Cov}(\mathbf{1}\{\tau_1 > 1\}, \mathbf{1}\{\tau_2 > 1\}) = 0 - \frac{1}{2} \cdot \frac{1}{2} = -\frac{1}{4}.$$

Under independence this covariance equals 0: the anti-monotone coupling thus yields a strictly negative value.

Why this is impossible in the real ARF. Reusing the factorization from Section 1 and the law of total covariance

$$\text{Cov}(X, Y) = \mathbb{E}_S[\text{Cov}(X, Y \mid S)] + \text{Cov}(\mathbb{E}[X \mid S], \mathbb{E}[Y \mid S]),$$

with $X = \mathbf{1}\{\tau_1 > 1\}$, $Y = \mathbf{1}\{\tau_2 > 1\}$: the first term is zero (conditional independence given S , Section 1). The second is computed directly, since $\mathbb{E}[X \mid S] = \mathbb{E}[Y \mid S] = G_S(1)$:

$$\text{Cov}(\mathbf{1}\{\tau_1 > 1\}, \mathbf{1}\{\tau_2 > 1\}) = \text{Var}(G_S(1)) \geq 0 \quad \text{always.}$$

Conclusion — the trap defused. The real ARF can only produce a covariance ≥ 0 between two trees — never negative. The anti-monotone coupling, which requires a covariance of $-1/4$, is therefore structurally impossible in the real model: the dependence shared through S can only go one way (independence if $\text{Var}(G_S) = 0$, positive dependence otherwise), never the other. This confirms and locks in the conclusion of Section 2: no negative dependence scenario could contradict “independence is too optimistic about speed”.